



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
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Roll Number	2520090205
Branch	CSIT
Course Outcome	3

Case Study Topic: SecureVote – Digital Voting & Election Analytics System using BFS

Work Done Summary:

Implemented **Breadth First Search (BFS)** in Java for election monitoring and voter network traversal. Modeled voters, polling stations, and election activities as graph nodes connected through edges. BFS traversal was used to monitor voting activities level by level and detect suspicious voting patterns. Implemented voter registration, authentication verification, ballot casting, vote counting, election analytics, and fraud detection using graph traversal techniques. Demonstrated BFS traversal for auditing connected voting records and analyzing election flow efficiently. Compared BFS traversal with DFS – BFS is more suitable for level-wise monitoring and shortest-path based election analysis.

Implementation Details :

1. Created Graph class using adjacency list representation.
2. addEdge() connects voter nodes and polling stations.
3. BFS(startNode) uses Queue data structure for traversal.
4. Visited array prevents duplicate traversal of nodes.
5. BFS traversal monitors election activities level by level.
6. Fraud detection performed by identifying suspicious repeated connections.
7. Main method created voter graph and demonstrated BFS traversal.
8. BFS ensures efficient election auditing and monitoring operations.

Result : Screenshots/Output

```
PS D:\JavaPrograms> javac SecureVoteBFS.java
PS D:\JavaPrograms> java SecureVoteBFS
SecureVote - Digital Voting System
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Graph Connections:
0 <-> 1
0 <-> 2
1 <-> 3
1 <-> 4
2 <-> 5

BFS Traversal:
Visited Node: 0
Visited Node: 1
Visited Node: 2
Visited Node: 3
Visited Node: 4
Visited Node: 5

Fraud Detection:
Checking duplicate voting activity...
Node 4 verified.
No Fraudulent Activity Detected.
```

Time Complexity:

- BFS Traversal: $O(V + E)$
- Graph Construction: $O(E)$
- BFS is efficient for connected voter network analysis and election monitoring systems.

GitHub Link : https://github.com/anumolajhansi08-sketch/DSA-2_CO3

Student Signature	Faculty Signature